MSC ARTIFICIAL INTELLIGENCE DISSERTATION

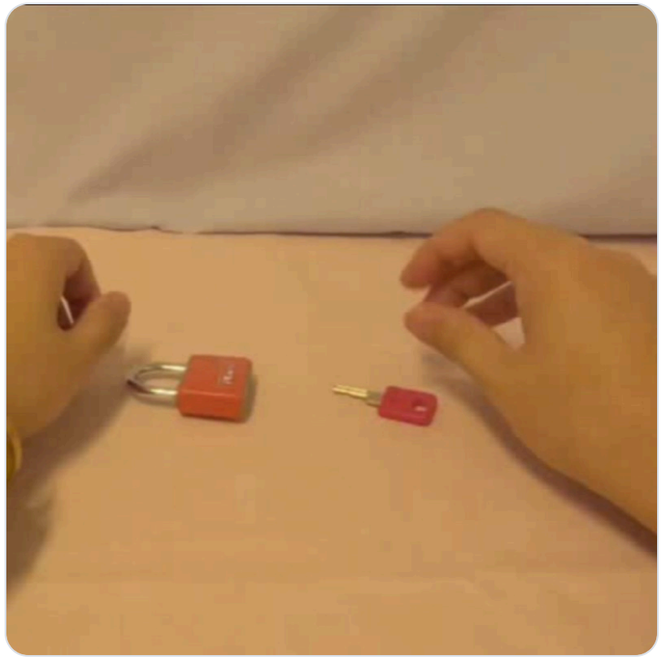
Learning to Select Actions for Complex Hand Tasks

A comparative study of visuomotor policies and latent world models

Graham Pellegrini
Supervised by Prof. Alexiei Dingli
Department of Artificial Intelligence, Faculty of ICT, University of Malta

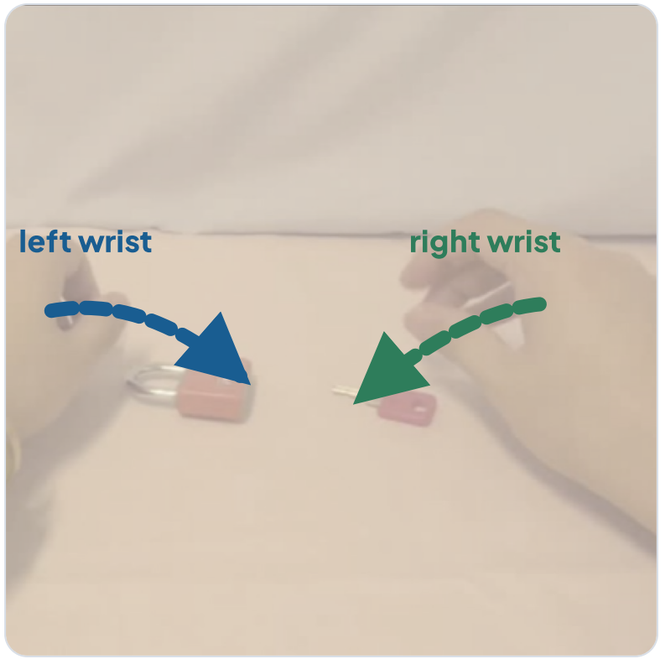
What is a policy?

A policy is a model that learns from demonstrations (human, robot or simulated) to turn the current scene into a chunk of future actions.



In this study, input: the current scene

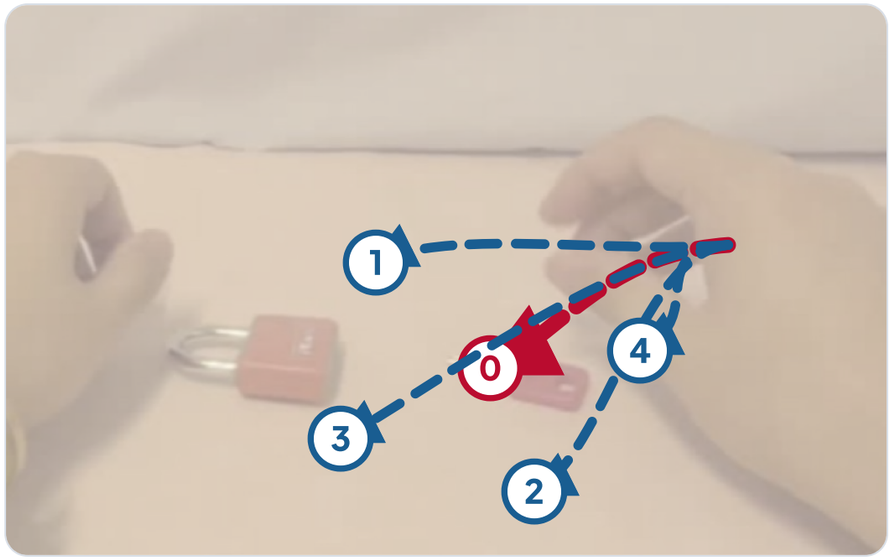




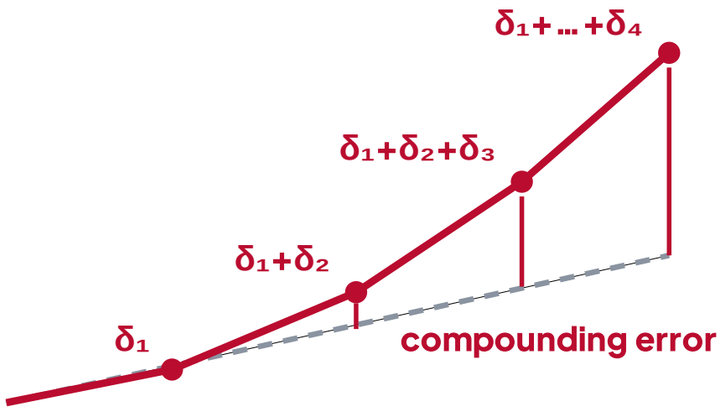
In this study, output: an action chunk, the next 16 wrist poses for both hands (18 numbers per step)

WHERE A POLICY FALLS SHORT

A policy can produce many motions, but commits to one



Different latent or noise samples give different candidates.
Candidate 0 is the policy's default output: z = 0 for ACT, the first sample for Diffusion Policy.



expert motion (dashed) policy rollout

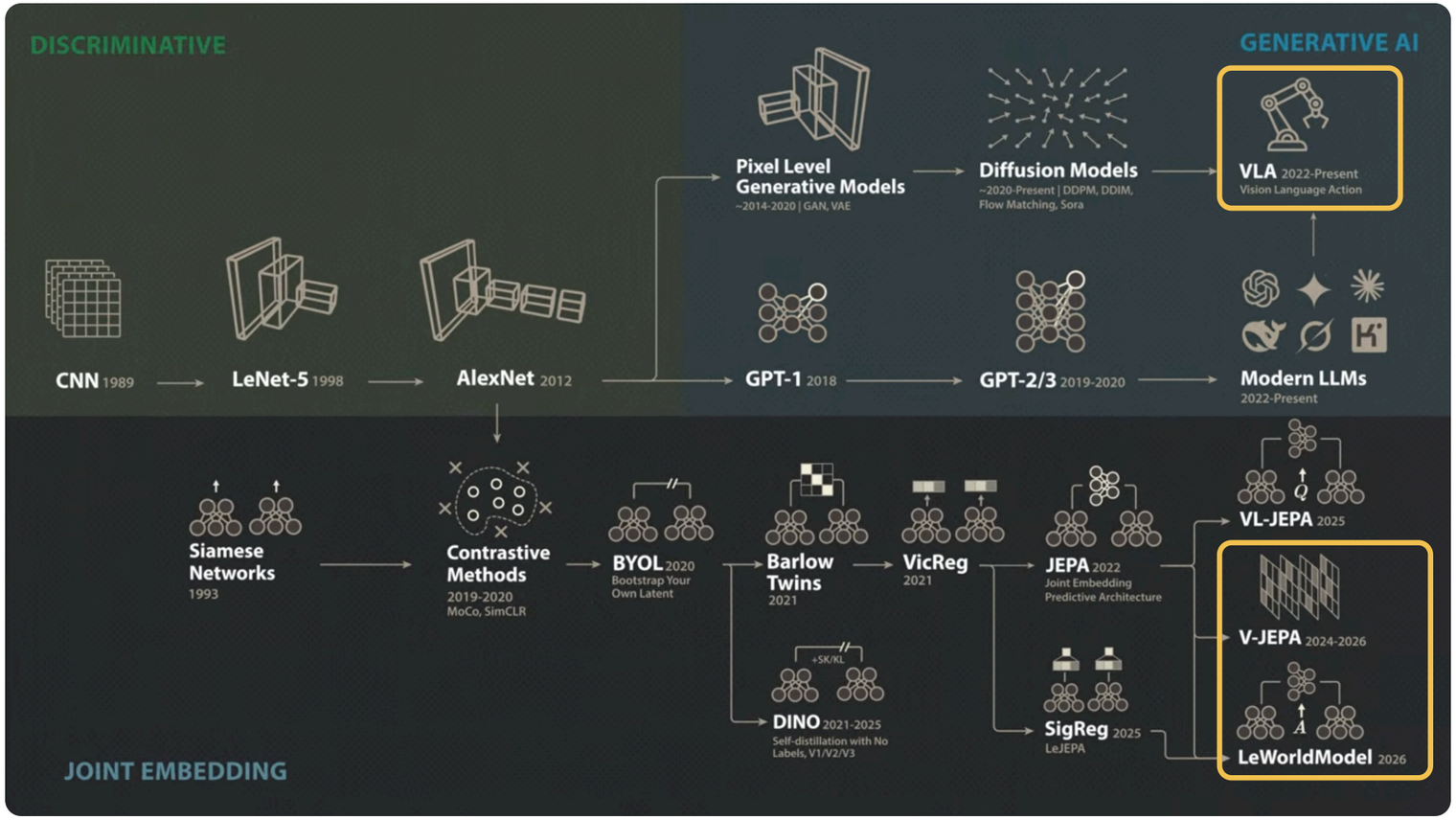
$$e_t = e_{t-1} + \delta_t = \delta_1 + \ldots + \delta_t$$

Suboptimal candidate → compounding error → unsuccessful or dangerous state

TECHNOLOGY OVERVIEW

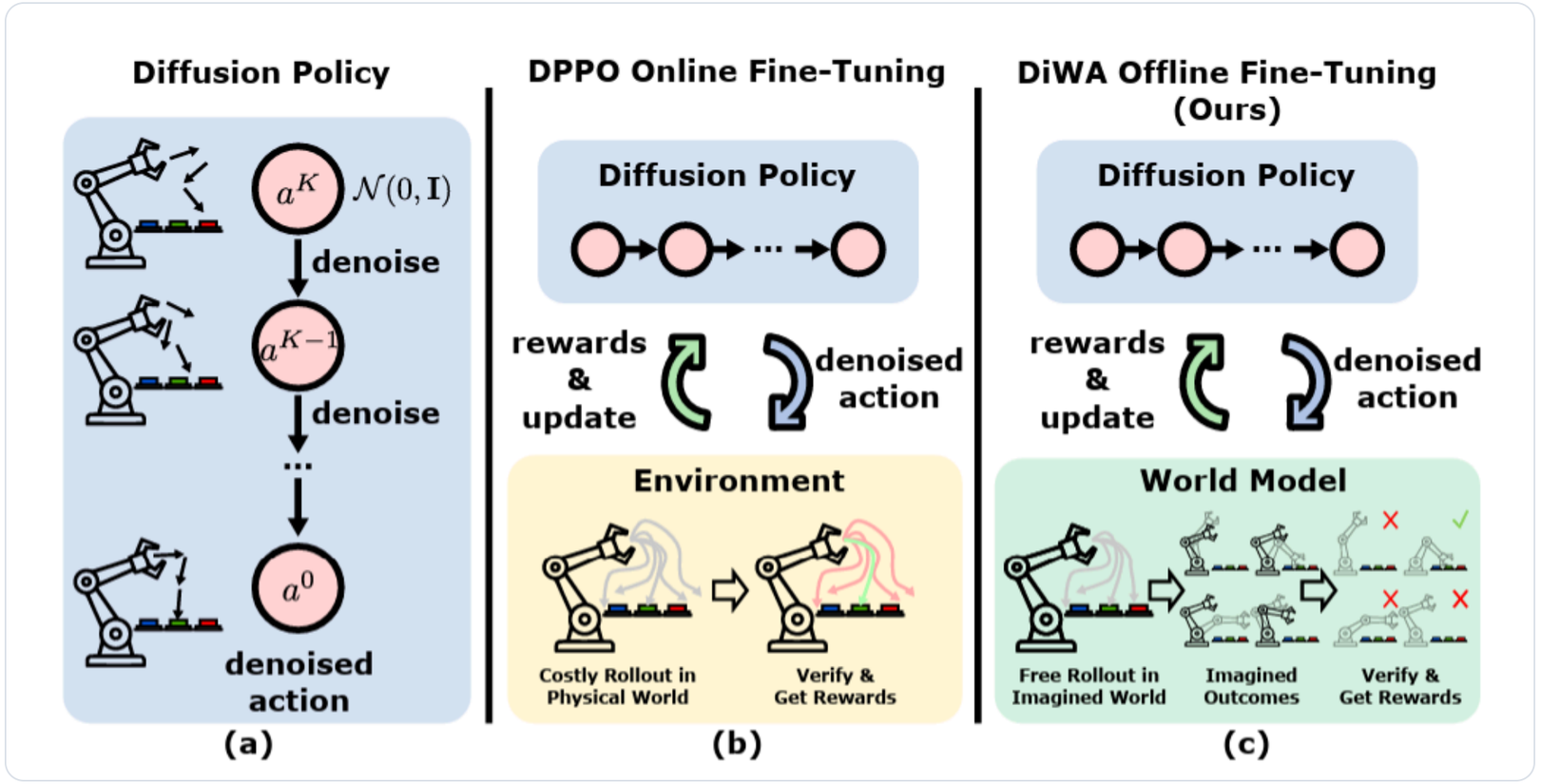
Why world models, not a VLA

A VLA generates the action directly; a world model predicts the consequence of an action, so an action can be checked before it runs.



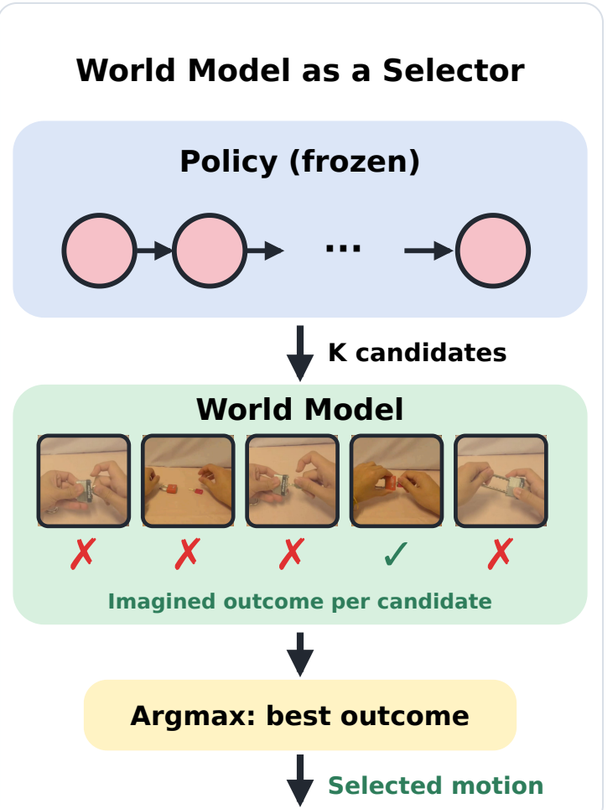
Screenshot from Welch Labs, Yann LeCun's $1B Bet Against LLMs (2026). Highlights added.

DIWA: THE PAPER WE BUILD ON



Left: Chandra et al. (2025), evaluated on CALVIN. This study reproduces (c). Right: illustrative frames.

OUR APPROACH: EXTERNAL WORLD-MODEL SELECTION



RESEARCH QUESTION

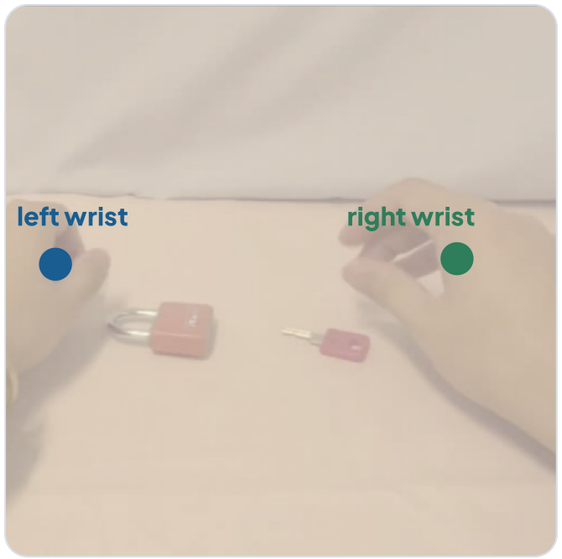
Can a frozen world model, used externally, select better motions from a policy's own candidates?

Tested offline, with the world model acting only as the final decision stage over candidates the policy already proposed.

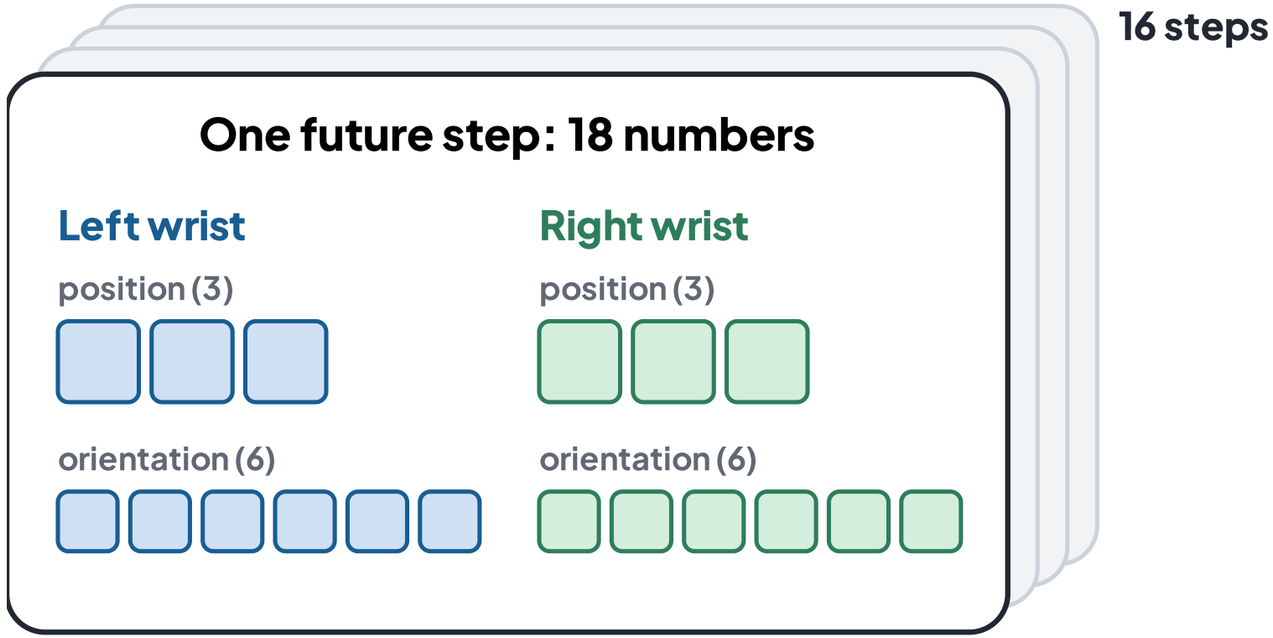
- How much does it help, and for which policy and world-model pairings?
- When does selection fail, and why?
- How does selecting compare with finetuning the policy inside a world model (DiWA on CALVIN)?

THE DATA

First-person video in, 18 numbers per step out



Egocentric video, 16 observed frames, both wrists tracked



18-D action per step: 9 numbers per wrist (3 position + 6 orientation).

6 DoF per wrist. The 6 orientation numbers encode 3 rotational DoF.

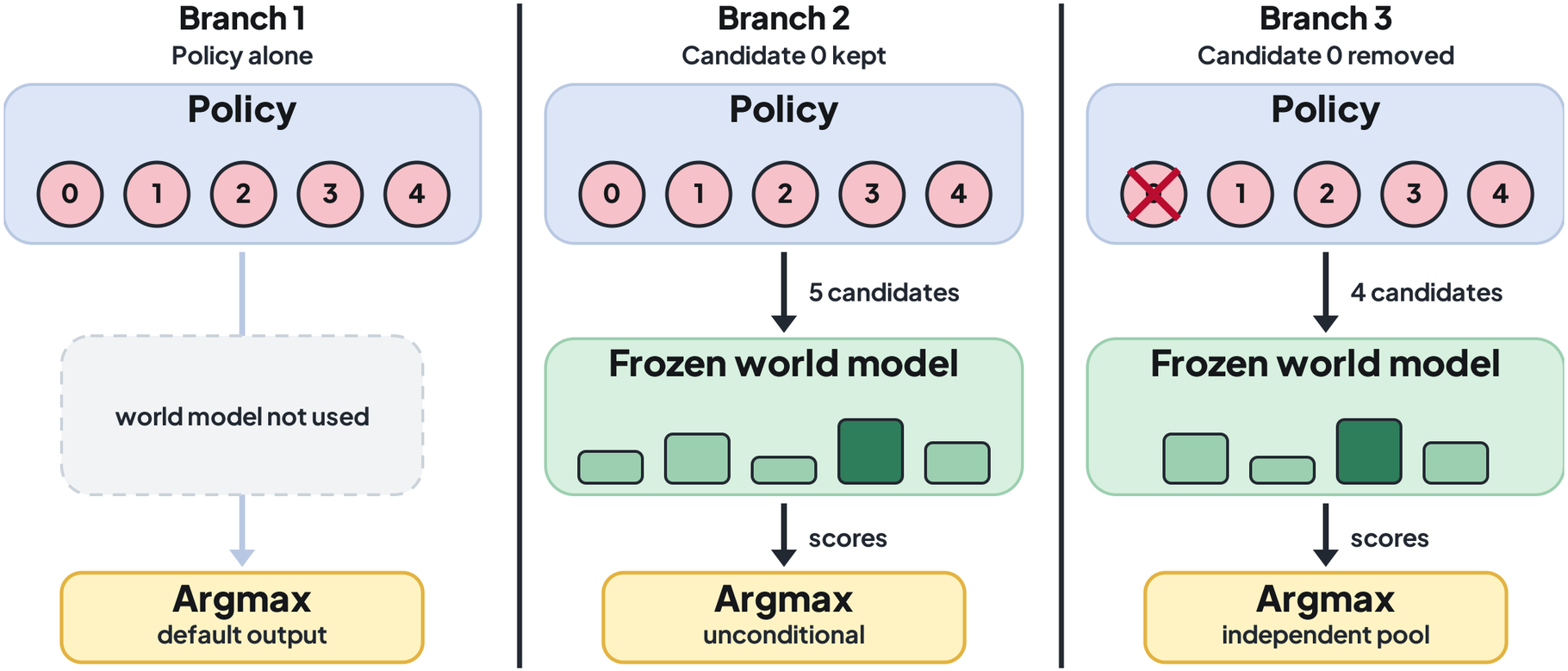
EgoDex
HUMAN BIMANUAL MANIPULATION, 30 FRAMES PER SECOND

16 in, 16 out
FRAMES OBSERVED, STEPS PREDICTED PER WINDOW

7,607 windows
HELD OUT, FROM 3,229 EPISODES AND 111 TASKS

THE METHOD

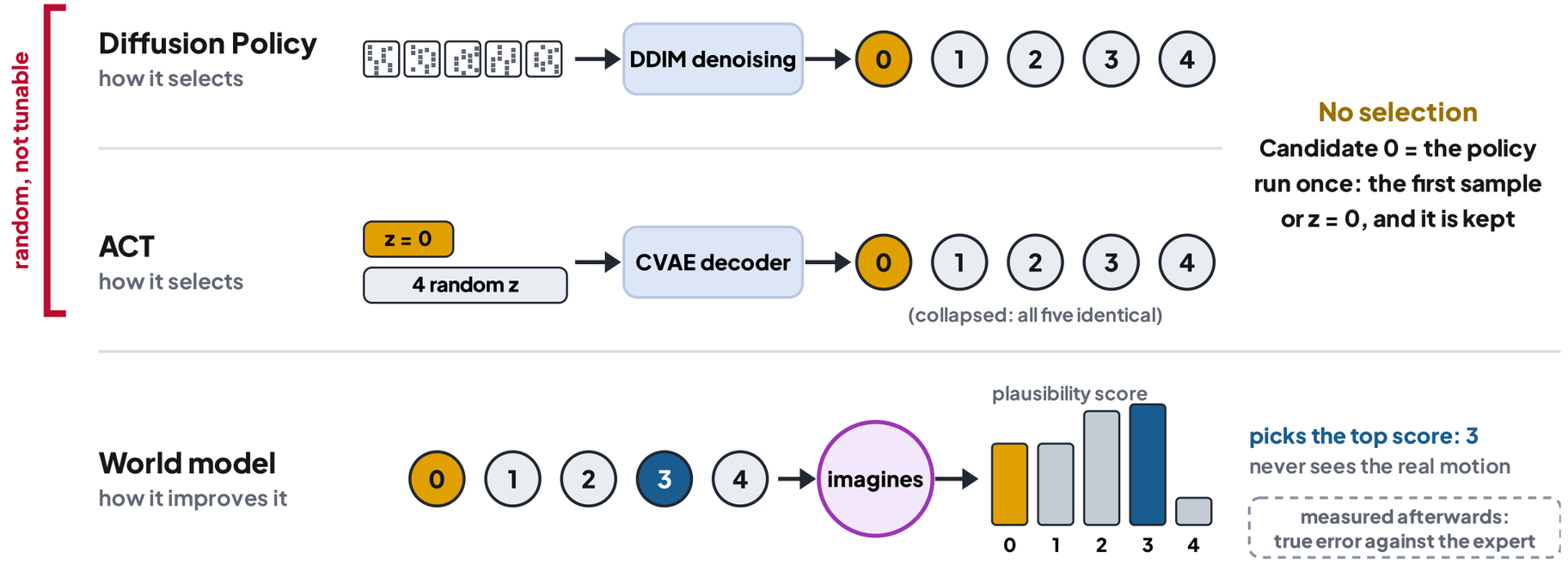
One pool of candidates, three ways to choose



7,607	16 to 16	18-D	K = 5	3 seeds
HELD-OUT WINDOWS	FRAMES IN, FRAMES OUT	BIMANUAL WRIST MOTION	CANDIDATES PER WINDOW	POOLED RESULTS

WHERE CANDIDATE 0 COMES FROM, AND WHAT THE WORLD MODEL ADDS

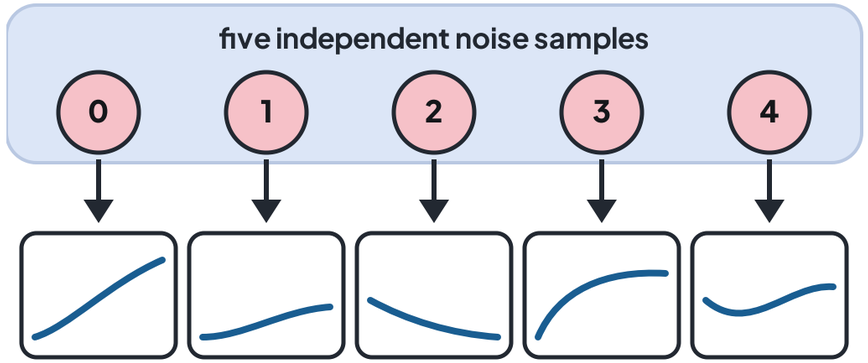
How each candidate gets chosen



WHERE THE CANDIDATES COME FROM

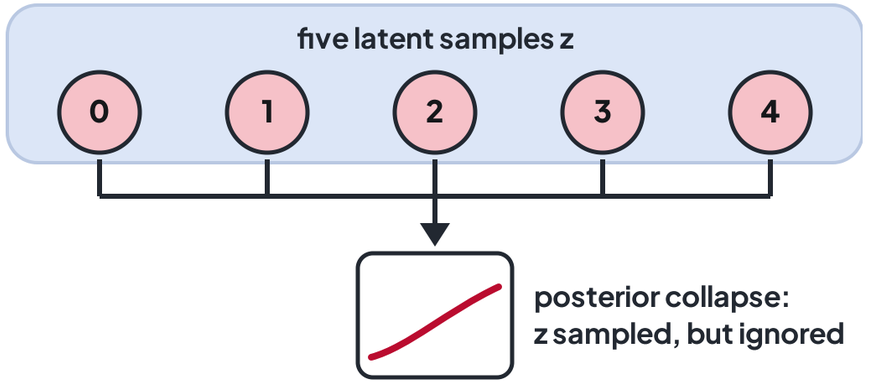
The two policies: Diffusion Policy and ACT

Diffusion Policy



Five genuinely different motions
independent pool, clean measurement

ACT



Five copies of one motion
no diversity, so selection has nothing to choose between

Diffusion Policy denoises Gaussian noise into a motion over DDIM steps. 5 independent noise samples give 5 candidates.

ACT is a CVAE, where a latent z sampled at test time should vary the motion. Posterior collapse means the decoder learned to ignore z, so all 5 samples are the same motion. That is a failure, not a goal.

ACT: four latent training configurations

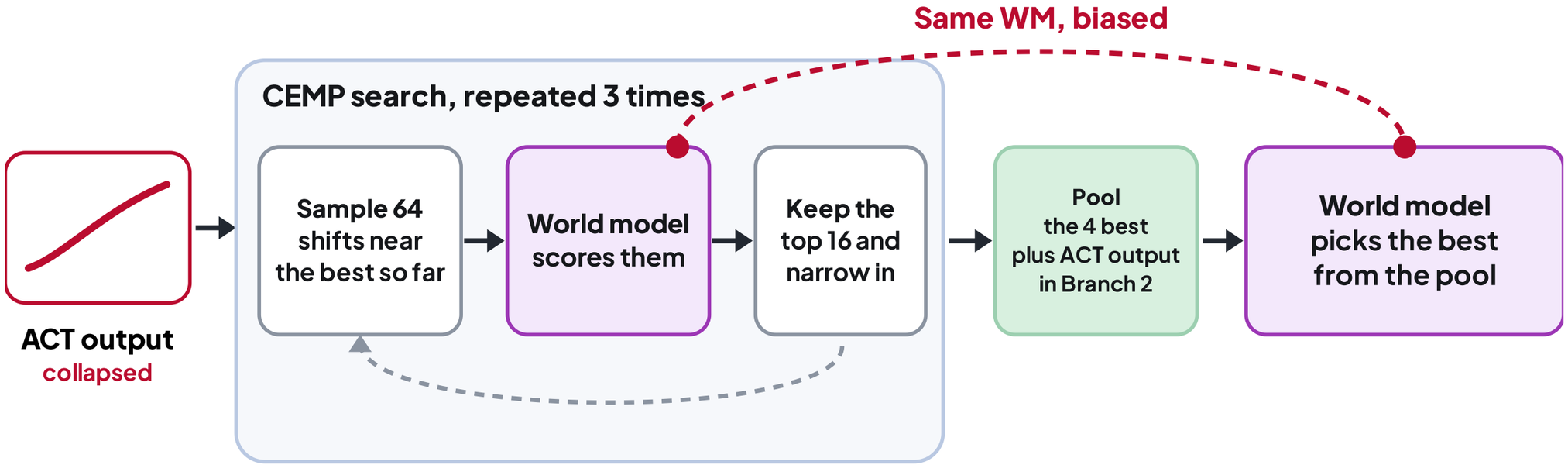
RUN	RECIPE	WHAT IT IS MEANT TO DO
v1	KL weight 10, the standard recipe	Baseline: the decoder can ignore z
v2	KL weight 100	Penalise information in z ten times harder
v3	Cyclical KL annealing and 30% observation dropout	Ease the penalty in cycles and hide observations, so the decoder must lean on z
v4	Cyclical KL annealing and free bits, 0.5 nats per dimension	Let z carry a minimum amount of information for free

No diversity

- All five samples identical
- Differences below 10^{-6}
- Posterior collapse

CEMP: Cross-Entropy Method with Policy

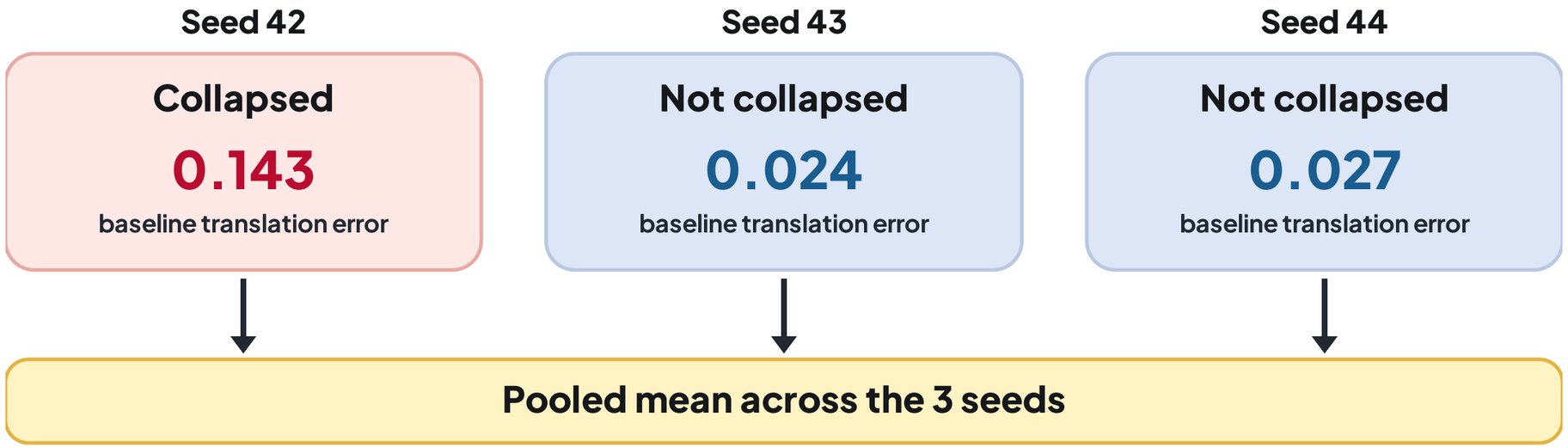
An adaptation of the Cross-Entropy Method initialised from the policy's output rather than a uniform prior, inspired by the policy-guided model-based planning strategy used in TD-MPC2. Adapted, not a published recipe.



- The world model shapes the pool, then picks from it, so the pool is biased towards its own taste.
- ACT results therefore measure the world model and the planner as one unit.

WHERE CEMP WAS APPLIED

CEMP was used only on the seed that collapsed



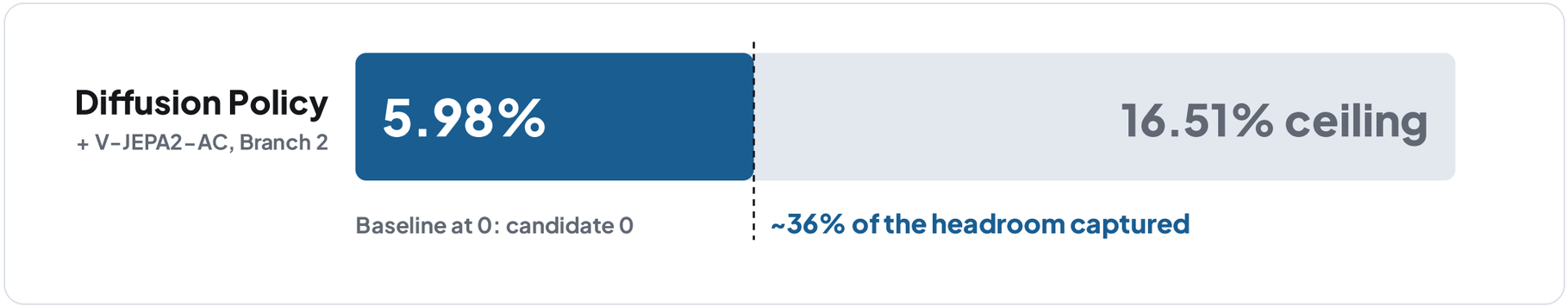
Only seed 42 needed CEMP, so seeds 43 and 44 were scored on ACT's own candidates. The pooled ACT result therefore mixes a world-model-plus-planner measurement with a plain scorer measurement. Read it as one result for this rule, not as a clean measurement of the scorer. This is why Diffusion Policy carries the main claim.

Five world-model families, one validated gain

MODEL	WHAT IT IS	WHAT IT SHOWED
V-JEPA2-AC	Pretrained video model	-5.98% the validated gain
DexWM	Dexterous-hand model	+0.57% no gain, one seed
LeWM	Small, from scratch	-10.1% fails ranking gate
DINO-WM, JEPA-WM	Official 7-D models	latent only interface mismatch
No-AC, PointWorld	Action cut or weakened	0.00%, +0.27% action is the mechanism

Translation error change, Diffusion Policy, Branch 2. Matched local exposure of 100,000 items. LeWM's score does not predict the better candidate, so it is held out (Backup C). ACT with CEMP gave -0.58%, read as one unit (slides 11 to 13).

Ranking Diffusion Policy candidates with V-JEPA2-AC cuts translation error by 5.98%

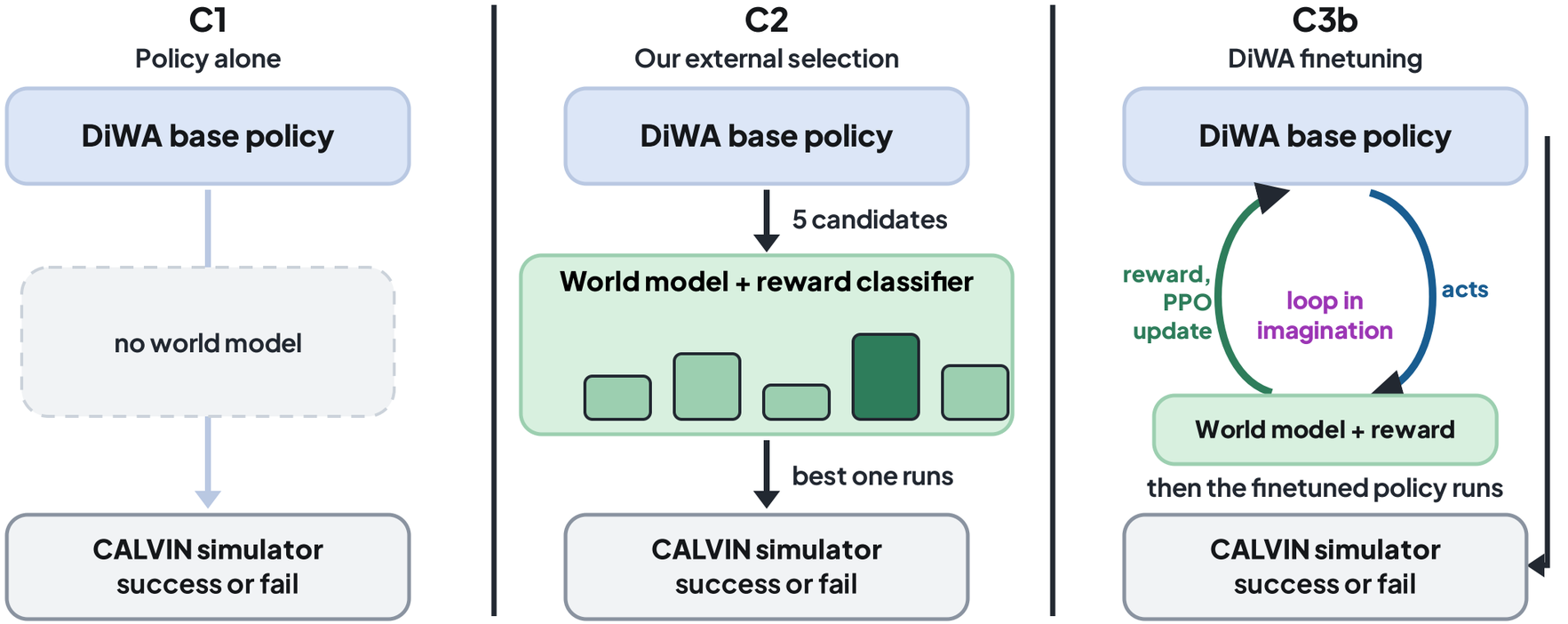


0.00% **Control: same scorer with the action input removed**
It cannot tell candidates apart, so every score ties. This tests the scorer, it is not the baseline.

The gain comes from reading each candidate's motion, not from the video backbone. Pooled across three seeds, 95% CI [5.43, 6.47], no retraining. Rotation error -0.96°. Win / tie / loss 49.9 / 23.2 / 27.0%. Branch 3: 5.37% of a 14.68% ceiling. The ceiling is what a perfect selector could remove from the same five candidates.

SECONDARY STUDY ON DIWA'S OWN BENCHMARK

Same policy and world model, three ways to use them



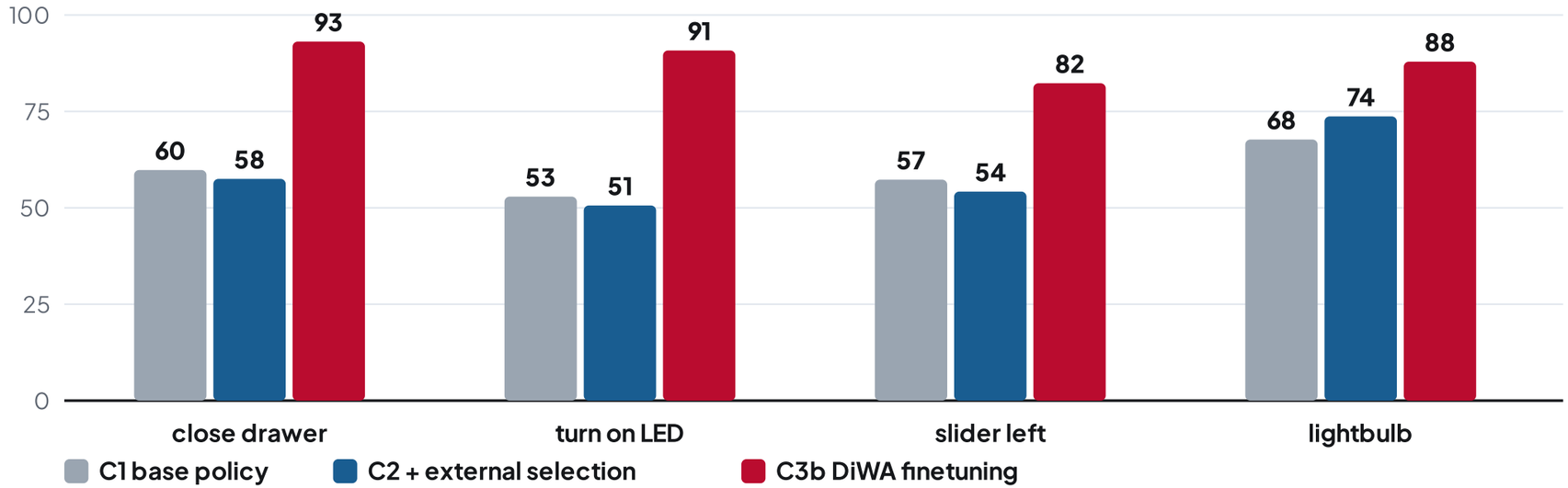
4 tasks
CALVIN MANIPULATION SKILLS

~100 rollouts
PER TASK, POOLED

3 seeds
42, 43 AND 44

Success rate
% OF EPISODES COMPLETED

Finetuning the policy in a world model beats selecting with one



SELECTION (C2 VS C1)

-3.1 to +6.1 points, no measurable effect.

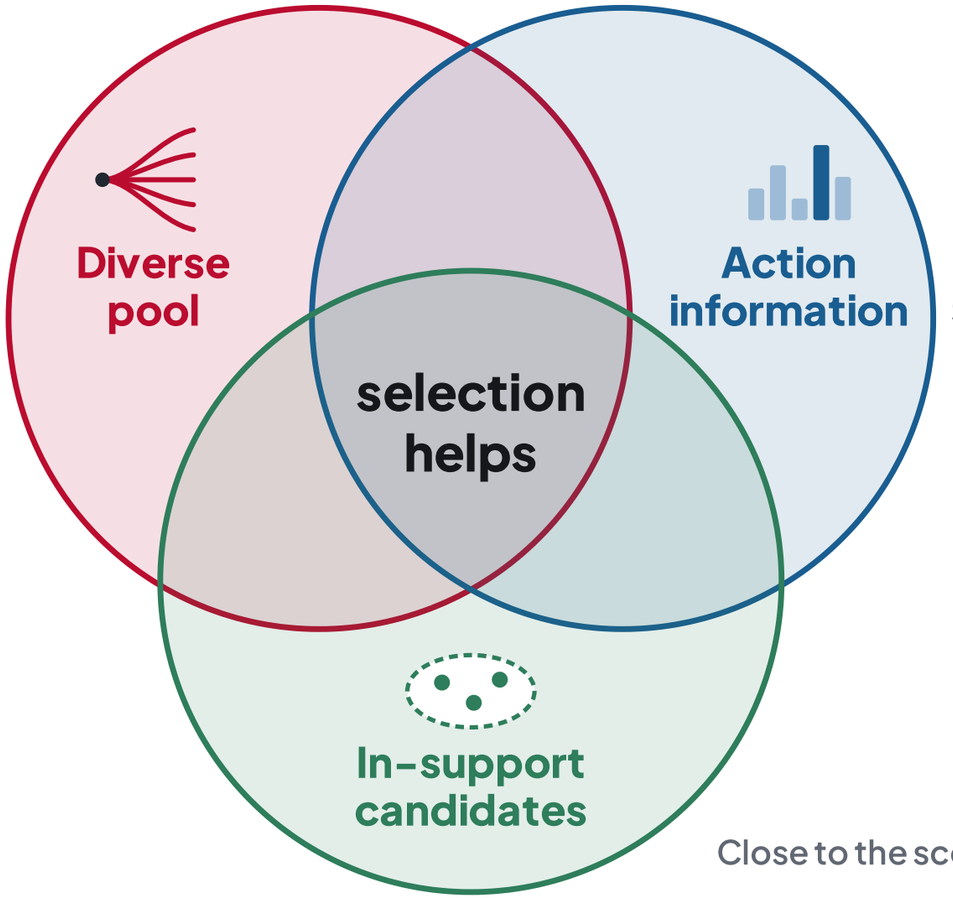
FINETUNING (C3B VS C2)

+14 to +40 points on every task.

Success is all or nothing, so a better pick only counts if it flips a whole episode. Selection's intervals overlap C1's on every task. C3b is read on DiWA's own evaluation loop.

What a world model needs to help

Different motions



Score follows the motion

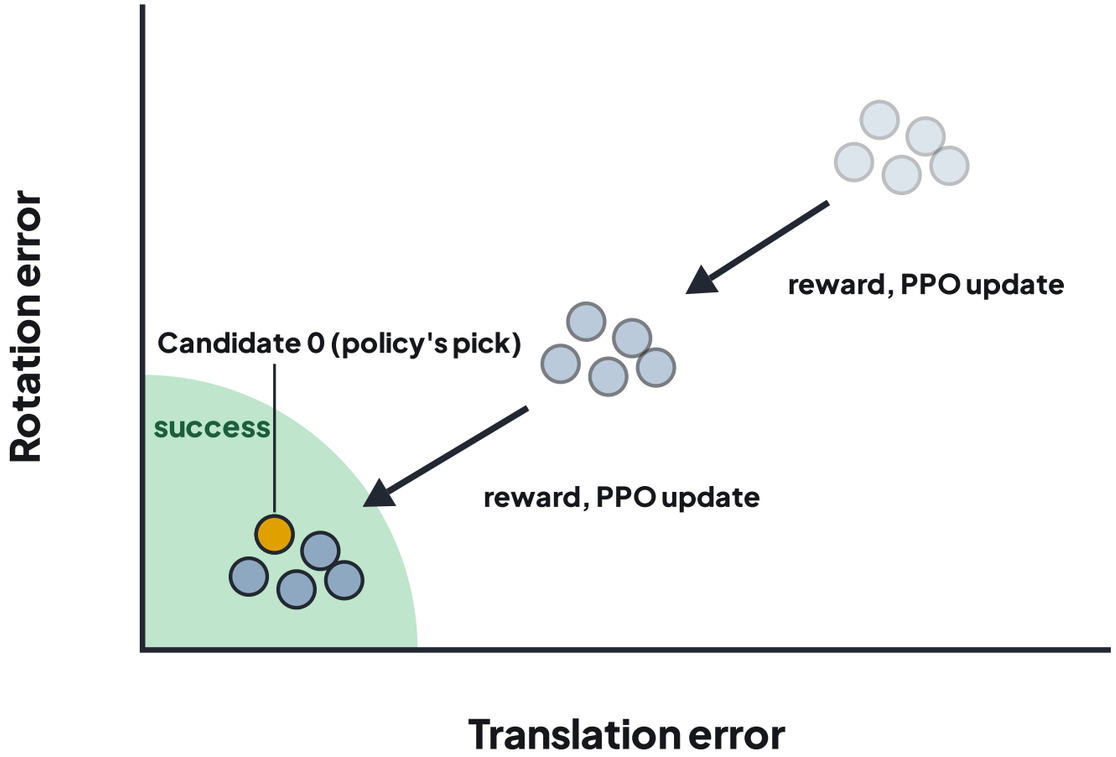
Close to the scorer's training data

Main pairing (DP + V-JEPA2-AC): all three

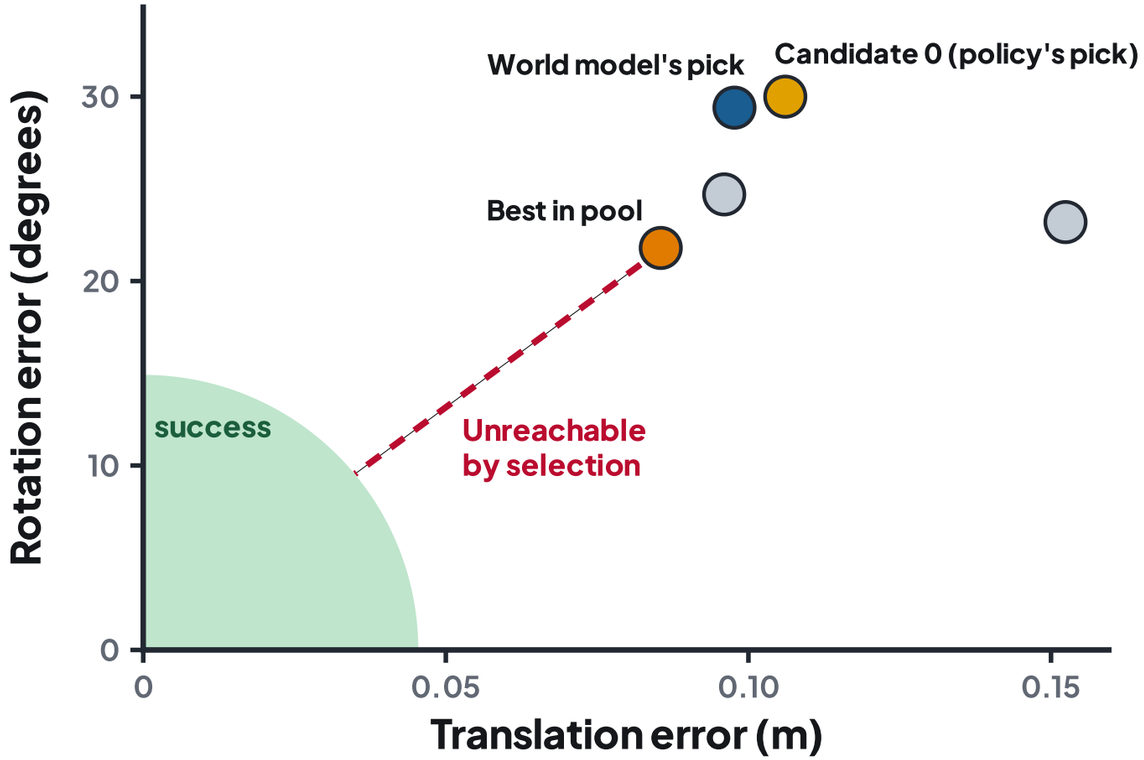
A pattern from the tested configurations, not a general law.

Finetuning moves the pool; selection can only pick from it

DIWA (ILLUSTRATIVE)



OUR SELECTION



Finetuning improves the policy. Selection improves the choice.

What this does and does not show

LIMITS

- Offline trajectory error, not robot task success
- One dataset and one positive pairing
- Selection worsens the easiest fifth of windows

NEXT STEPS

- Closed-loop execution on CALVIN-style tasks
- A world model as selector over a finetuned policy
- Native 18-D finetuning of DINO-WM and JEPA-WM

Takeaways

It can help

On one pairing, a frozen world model cut translation error by 5.98% with no retraining.

It has a ceiling

Selection took ~36% of a 16.51% ceiling. The rest needs a better pool.

It needs the right pairing

Diverse pool, action information and in-support candidates, all together.

> **A frozen world model can improve which of a policy's own motions is chosen, without retraining, for one tested pairing.**

NEXT, IN M.AI.ESTRO

18-D wrist poses → robot motion → closed loop on a robot

THANK YOU FOR LISTENING

Questions

Backup slides follow, covering the AI-use workflow, how each method was chosen, LeWM's gate, seeds, pool size, and where selection fails.

Graham Pellegrini | graham.pellegrini@um.edu.mt

How AI was used in this work

NotebookLM

Queried paper sets for related work and citations, then each source was skimmed manually.

Claude skills

Query, draft and polish workflows, with my own review and edits.

Gemini

Paper discovery and critical review of chapters against a rubric.

Coding agents

Wrote most of the code under my specification, then tested and audited.

WHERE THE DECISIONS SAT

Research question, benchmark design, which methods to adopt, what counted as a valid result, and every claim in the dissertation.

HOW CORRECTNESS WAS CHECKED

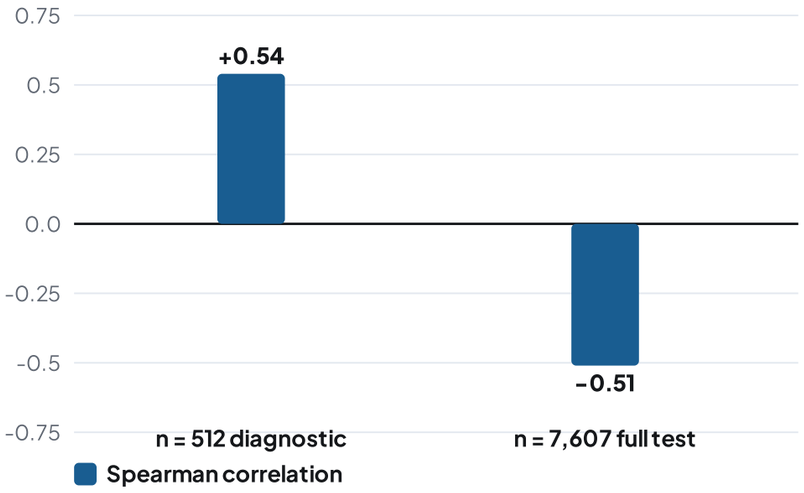
Separate audit passes and re-runs found real defects, including a DDIM sampler indexing error, a training-exposure undershoot, and a DiWA configuration mistake. All were fixed before reporting.

Used most in the literature review and methodology, for example to find established reference methods for handling ACT and LeWM without inventing new verification tests.

Which published work anchored each method choice

CHOICE	ANCHOR	STATUS
Policy-guided candidate search (CEMP)	Cross-entropy method, initialised from the policy output, inspired by TD-MPC2	Adapted, not a published recipe
Posterior-collapse fixes for ACT	KL annealing (Fu et al.), free bits (Kingma et al.), observation dropout	All tried, all failed
Diffusion sampling	DDIM (Song et al.), DP-T architecture (Chi et al.)	Implemented, sampler bug fixed
Action-conditioned scorer	V-JEPA2-AC (Assran et al.)	Head trained locally
Finetuning comparison	DiWA (Chandra et al.) on CALVIN, own checkpoints	Reproduced, secondary study
Pool size K = 5	No literature anchor	Compute choice, ranking stable K = 2 to 20
Matched exposure 100,000 items	Compute ceiling, loss still falling	Not a convergence claim

LeWM passes its ranking gate on 512 windows and fails it on 7,607



WHAT IS AND IS NOT ESTABLISHED

- The gain is real, −10.1% translation, intervals exclude zero
- The reported configuration never invokes LeWM's learned dynamics
- Its score does not predict which candidate is better at full scale
- Reweighting terms to pass the gate was rejected as overfitting the gate

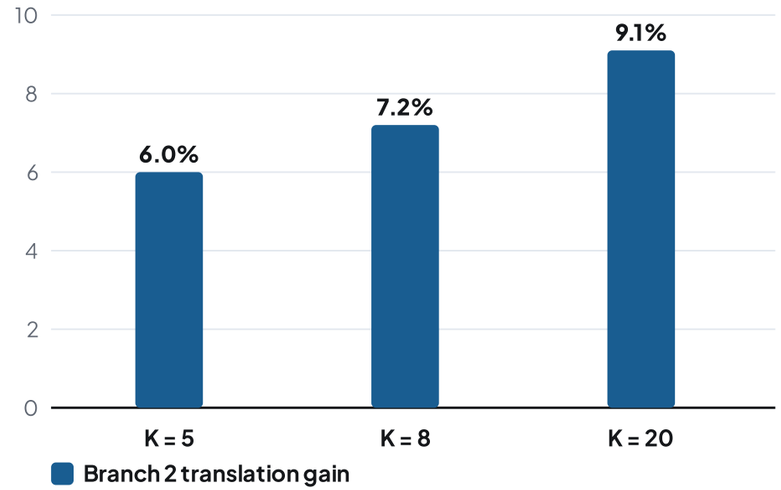
Spearman correlation between score margin and true translation improvement. The 512-window figure was +0.537, the full test set gave −0.51.

The result holds at every seed, and the controls behave as designed

	SEED 42	SEED 43	SEED 44
Baseline translation error	0.1142	0.1178	0.1191
V-JEPA2-AC, Branch 2	-6.41%	-6.09%	-5.46%
V-JEPA2 without action input	0.00%	0.00%	0.00%
PointWorld proxy	+0.50%	+0.23%	+0.07%
LeWM rollout (unvalidated)	-10.10%	-10.06%	-10.12%

Three seeds are a basic robustness check, not proof of stability. DexWM is single-seed by a cost decision of roughly 121 extra GPU-hours.

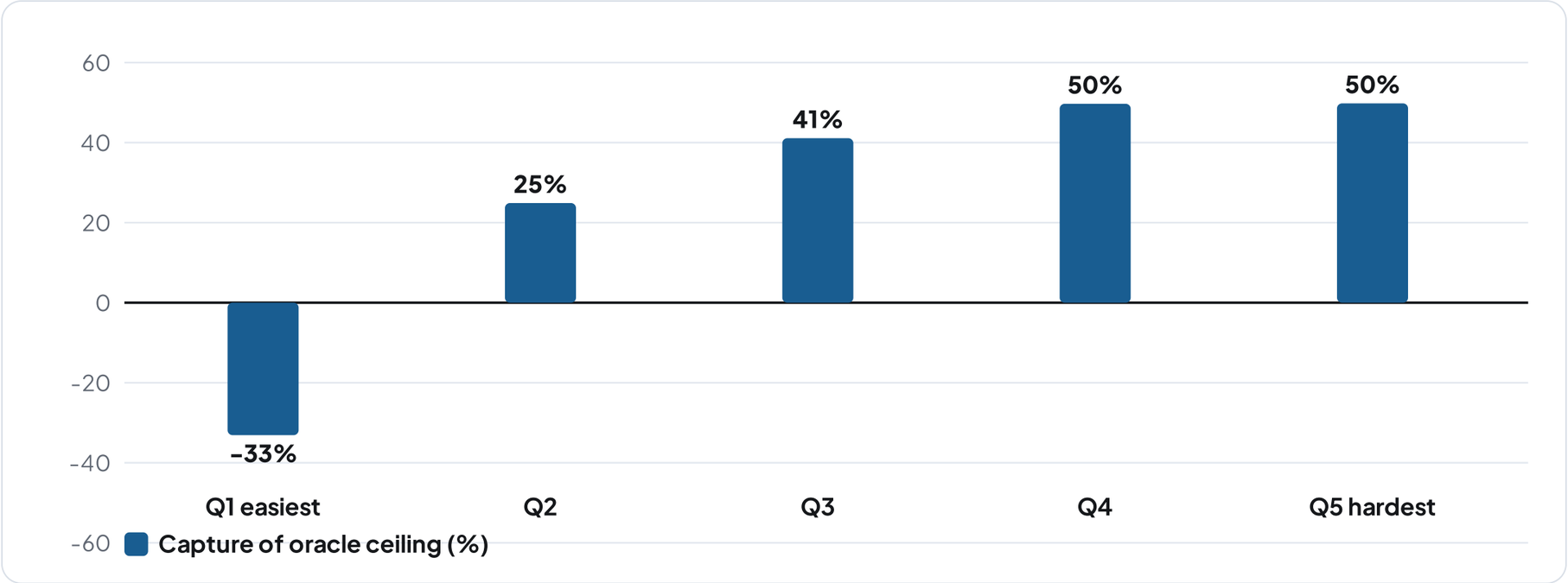
More candidates buy more headroom, not better selection



WHAT THE SWEEP SHOWS

- Scorer ranking identical at every K from 2 to 20
- Gain grows steadily with K and does not plateau
- Capture of the oracle ceiling stays flat, near 40% on these nested pools
- K = 5 was a compute choice, with no literature anchor

On the easiest fifth of windows, selection makes things worse



Windows sorted by baseline error, K = 8, Branch 2. Translation error rises 5.81% in Q1. Confidence gating does not help, because the score gap has a correlation of -0.002 with baseline error. A state-change-targeted windowing removes most of the reversal.

How results were measured

ERROR METRICS

- Translation, root-mean-square per axis, not Euclidean distance
- Rotation, mean geodesic angle in degrees
- Both averaged over the full 16-step chunk
- Change reported relative to candidate 0

STATISTICS

- 95% episode-clustered bootstrap, 2,000 resamples, pooled over seeds
- Oracle ceiling, best candidate in the same pool
- Ranking-validity gate, per-window Spearman correlation
- Win, tie, loss against candidate 0

Test windows are fixed, candidate pools frozen, future-goal access blocked, and scorer support restricted to training data.

What is disclosed about test-set use

- The Branch 2 margin threshold's final value came from a disjoint validation split.
- An earlier, superseded calibration touched the test set directly. This is stated as a named limitation and is reassuring, not a full defence.
- The LeWM gate was tuned on a small diagnostic split and failed to generalise to the full test set. It was reported, not re-tuned.
- Earlier stronger-looking results relying on shortcuts, transductive support or oracle access were corrected rather than reported.

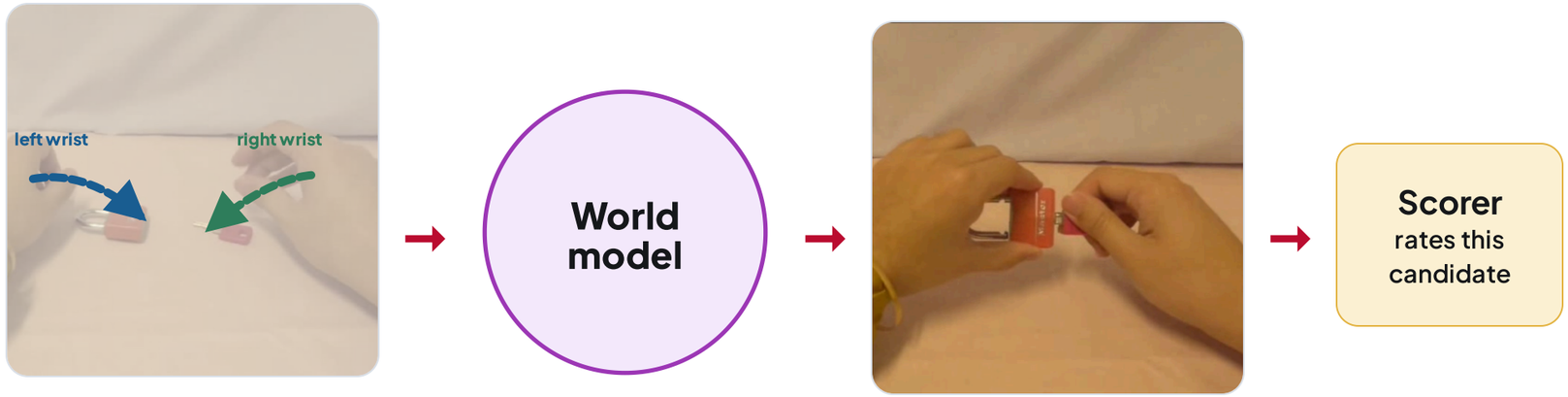
Objective achievement

OBJ.	STATUS	BASIS
1	Fully	Three-branch benchmark implemented as designed and used for every family reported
2	Partly	Local exposure matched, externally pretrained backbones carry an advantage not equalised
3	Within tested configurations	Whether and when selection helps determined for every combination, one positive, a likely explanation for each other

Objectives 4 and 5 are in the Conclusion chapter's objective table. Every non-positive-route explanation is the study's own interpretation, not independently proven.

THE PREDICTION UNIT

A world model predicts what happens next, and a scorer turns that into a choice



Input: the scene and one candidate motion

Predicted scene afterwards

The world model does not decide what to do. A prediction is not a decision, so a scorer must turn each predicted future into a score. Illustrative frames: the real prediction is made in a compact latent space, not in pixels.

Our CALVIN reproduction against the DiWA paper

TASK	PAPER BASE	OUR C1	PAPER DIWA	OUR C3B	OUR C2 (SELECTION)
close drawer	59.14	59.77	91.95	93.10	57.47
turn on LED	48.43	52.87	86.21	90.81	50.57
move slider left	62.15	57.29	83.33	82.29	54.17
turn on lightbulb	60.61	67.68	91.92	87.88	73.74

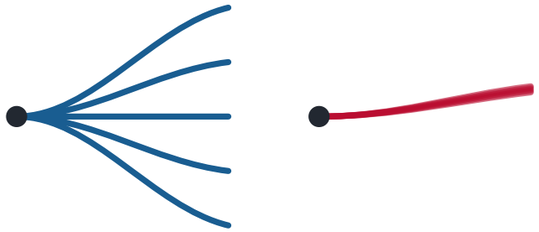
Success rate (%), three seeds. Paper values from Chandra et al. (2025), Table 1. Our runs use the authors' released checkpoints, world model and reward classifier. C1 and C2 share one evaluation harness; C3b is read on DiWA's PPO-trainer loop, so C1 and C3b come from different harnesses. Differences from the paper are within about 7 points and are expected from rerunning on different hardware and seeds. The behaviour-cloning weight α_{BC} was set to the paper's published values after an initial run at zero collapsed two tasks.

The three conditions, case by case

Diffusion Policy with V-JEPA2-AC met all three. Each red case is where one failed.

Diverse pool

✓ five different ✗ five copies



Diffusion Policy ACT, collapsed

The policy must offer genuinely different motions to choose from.

✓ Main pairing: met

Action information

✓ sees the motion ✗ blind to it



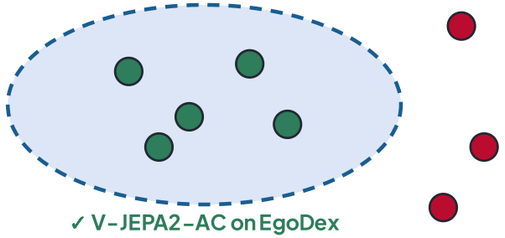
V-JEPA2-AC no-action control: 0.00%

The scorer's prediction must change with each candidate's motion.

✓ Main pairing: met

In-support candidates

what the scorer was trained on ✗ 7-D adapters



✓ V-JEPA2-AC on EgoDex

Candidates must look like the data the scorer learned from.

✓ Main pairing: met

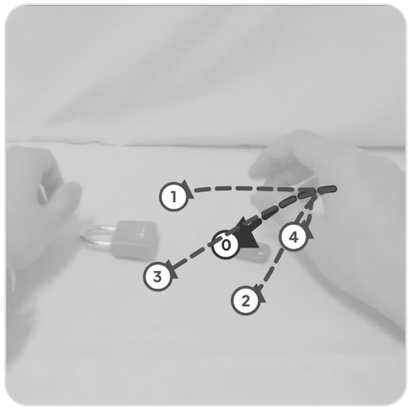
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DRAFT, not presented

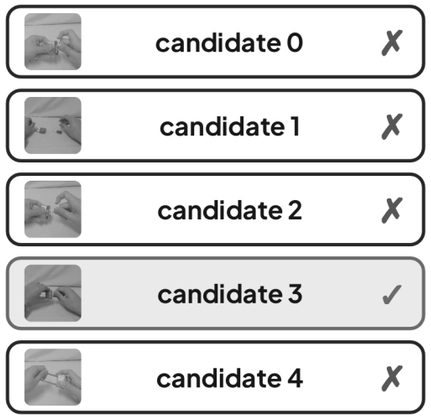
IMAGINE EACH OUTCOME, THEN CHOOSE THE BEST CANDIDATE

How our selection works



Five candidate motions

World model



Imagined outcome of each candidate

$$\arg\max_k s(\hat{o}_k)$$

picks candidate 3

compared with



The real outcome (expert motion)

The real outcome is used only to measure the error. The scorer never sees it.